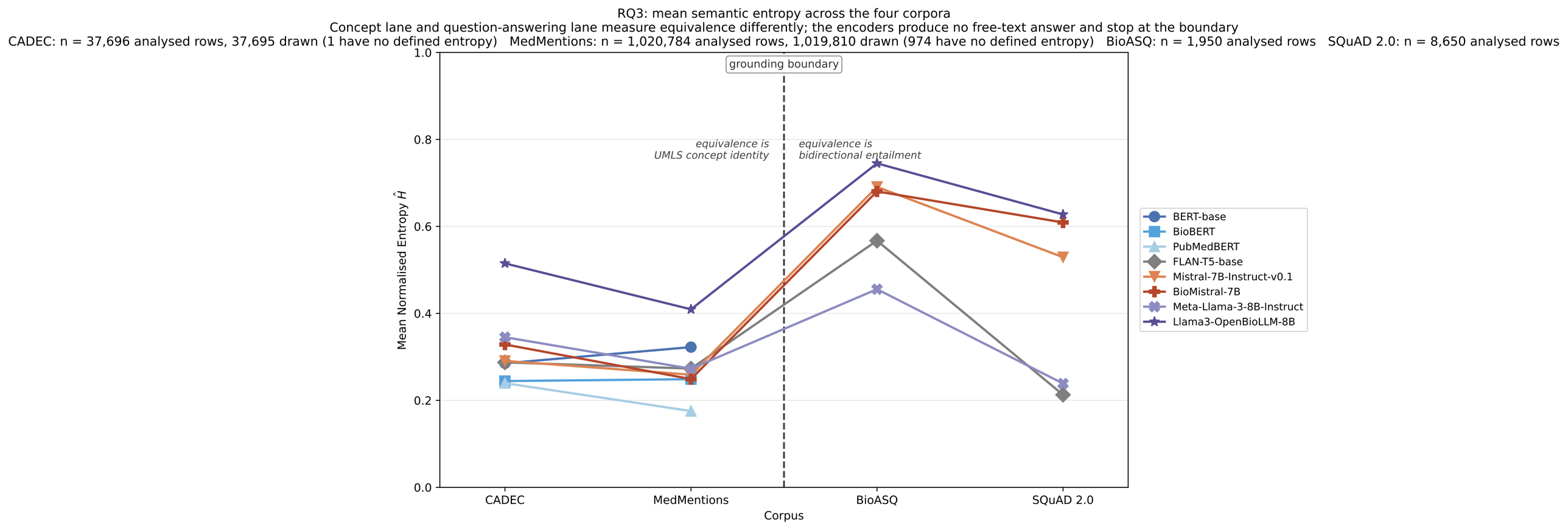
RQ3: mean semantic entropy across the four corpora
Concept lane and question-answering lane measure equivalence differently; the encoders produce no free-text answer and stop at the boundary
CADEC: n = 37,696 analysed rows, 37,695 drawn (1 have no defined entropy) MedMentions: n = 1,020,784 analysed rows, 1,019,810 drawn (974 have no defined entropy) BioASQ: n = 1,950 analysed rows SQuAD 2.0: n = 8,650 analysed rows
grounding boundary
equivalence is UMLS concept identity
equivalence is bidirectional entailment
Mean Normalised Entropy $\hat{H}$
1.0
0.8
0.6
0.4
0.2
0.0
CADEC
MedMentions
BioASQ
SQuAD 2.0
Corpus
BERT-base
BioBERT
PubMedBERT
FLAN-T5-base
Mistral-7B-Instruct-v0.1
BioMistral-7B
Meta-Llama-3-8B-Instruct
Llama3-OpenBioLLM-8B